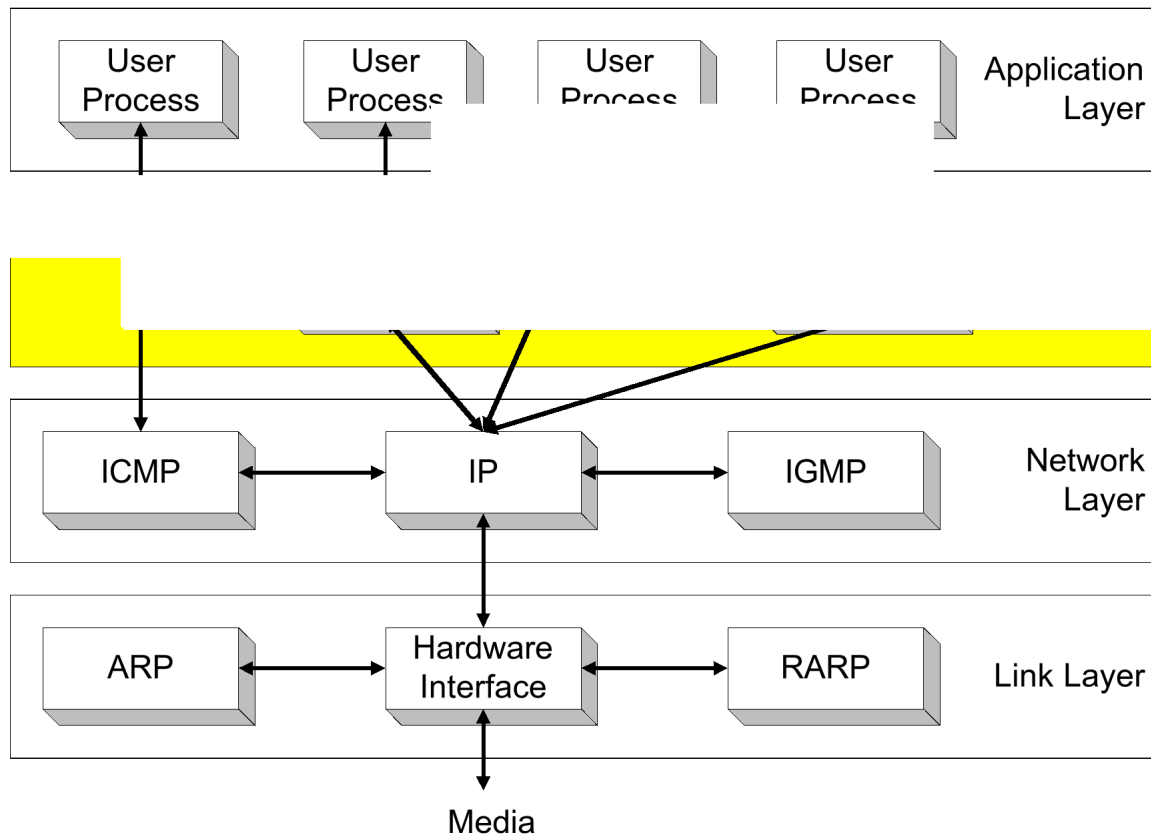

Relates to Lab 5. An overview of the transport protocols of the TCP/IP protocol suite. Also, a short discussion of UDP.

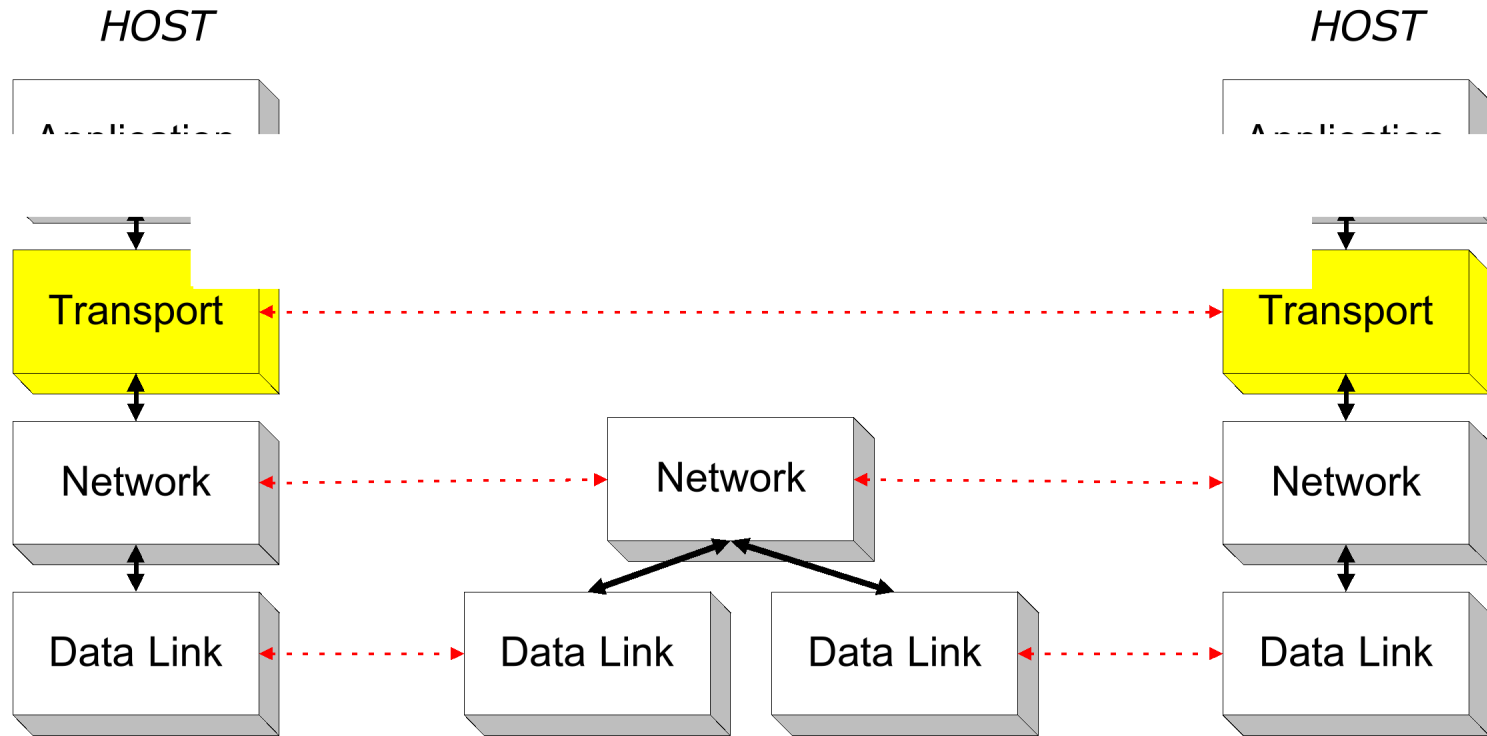
Orientation

- We move one layer up and look at the transport layer.



Orientation

- Transport layer protocols are end-to-end protocols
- They are only implemented at the hosts



Transport Protocols in the Internet

- The Internet supports 2 transport protocols

UDP - User Datagram Protocol

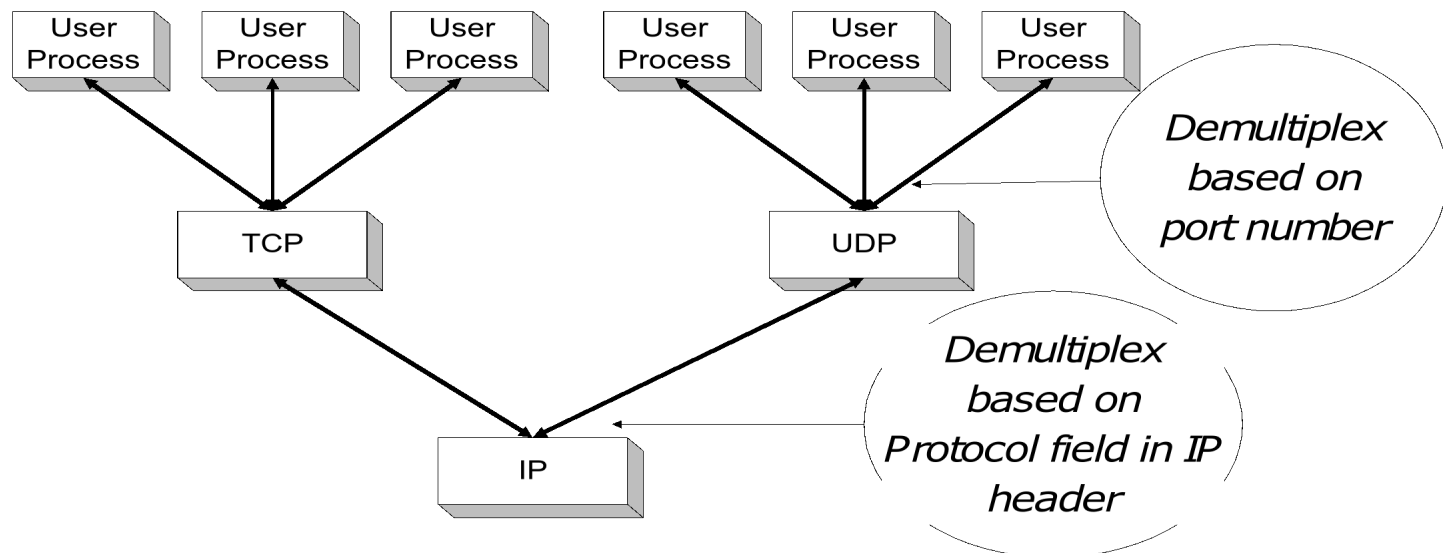
- datagram oriented
-
-
- unicast and
- useful only for few applications, e.g., multimedia applications
- used a lot for services
 - network management (SNMP), routing (RIP), naming (DNS), etc.

TCP - Transmission Control Protocol

- , full duplex
- reliable
- complex
- only unicast
- used for most Internet applications:
 - web (http), email (smtp), file transfer (ftp), terminal (telnet), etc.

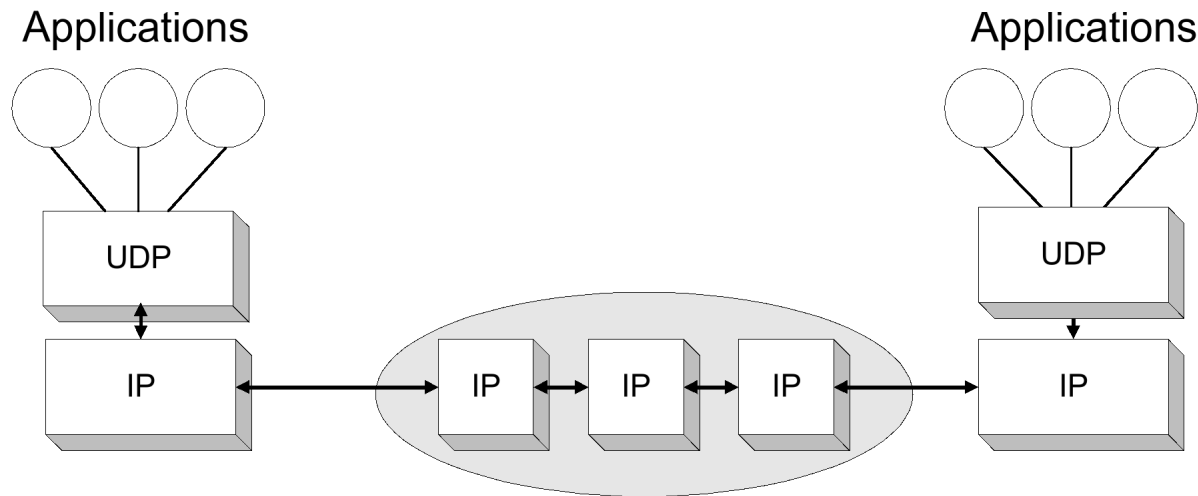
Port Numbers

- UDP and TCP use port numbers to identify applications
- A globally **unique** address at the transport layer (for both UDP and TCP) is the tuple **<IP address, port number>**
- There are $2^{16} = 65,536$ per protocol.
- S like email allocations. for random

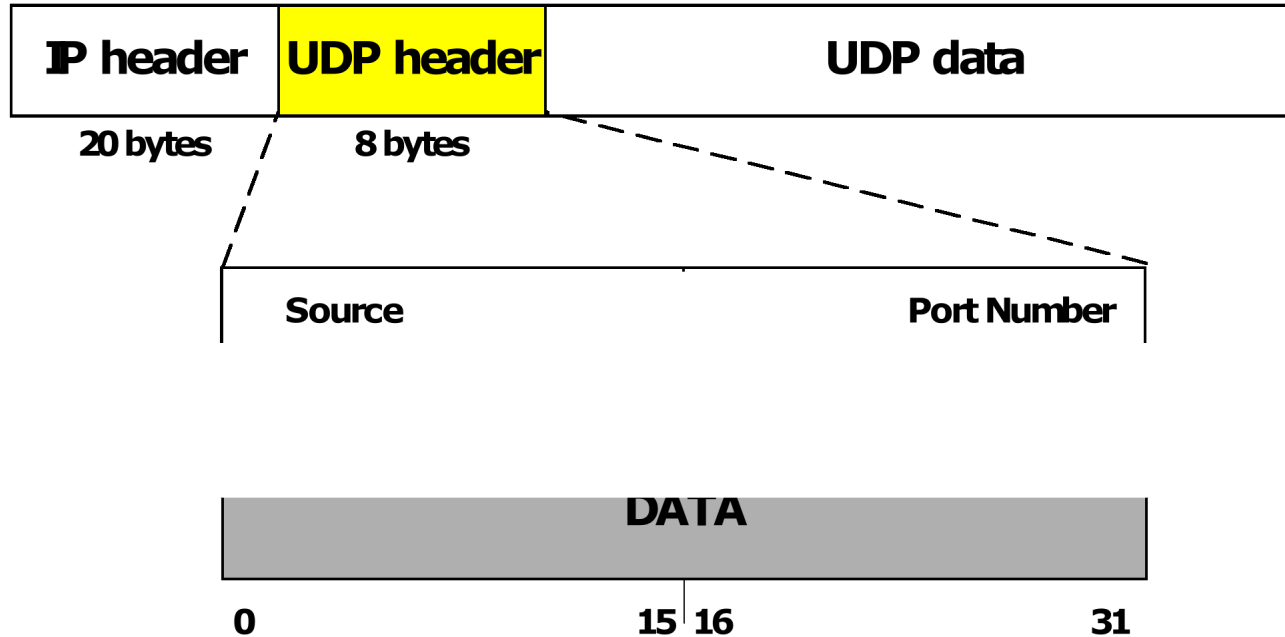


UDP - User Datagram Protocol

- UDP supports unreliable transmissions of datagrams
- UDP merely extends the host-to-host delivery service of IP datagram to an application-to-application service
- The only thing that UDP adds is multiplexing and demultiplexing



UDP Format



- **Port numbers** identify sending and receiving applications (processes). Maximum port number is $2^{16}-1= 65,535$
- **Message Length** is at least 8 bytes (i.e., Data field can be empty) and at most 65,535

UDP Checksum

- UDP checksum computation is optional for IPv4. If a checksum is not used it should be set to the value zero.
- The UDP checksum is calculated using a UDP-pseudo header, and **UDP packet** consisting of UDP header and the UDP data.
- The **UDP pseudo header** contains the source IP-address, the destination IP-address, the protocol identifier (UDP in this case), and the length taken from UDP header.

IP Destina

- Protocol T
- UDP length – header and data

IPv4 Pseudo Header Format

Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Source IPv4 Address																															
4	32	Destination IPv4 Address																															
8	64	Zeroes								Protocol								UDP Length															
12	96	Source Port																Destination Port															
16	128	Length																Checksum															
20	160+	Data																															